ApexPlanet Cybersecurity Internship

Task 2: Network Security & Scanning

Full Report

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Task	Task 2: Network Security & Scanning (Days 13-24)
LinkedIn	https://www.linkedin.com/in/rupashi-sethi-4505a936a/
GitHub repository	https://github.com/rupa-code/Task-2-Apexplanet-cybersecurity-intern
Video (LinkedIn post)	https://www.linkedin.com/feed/update/urn:li:activity:7519007174472368128/

Introduction

This report is my submission for Task 2 of the ApexPlanet Cybersecurity & Ethical Hacking Internship. The task is about reconnaissance, port and service scanning, and vulnerability scanning. I ran every scan on my own private lab: Kali Linux as the attacker and Metasploitable2 as the target. For the vulnerability scan I used Nessus Essentials, because OpenVAS did not run properly in my lab. All screenshots come from my own machines.

Task Deliverables

Deliverable	Where to find it
Nmap Scan Report	Section 3 of this report (Figures 5 and 6)
Vulnerability Report (Nessus Essentials, used instead of OpenVAS)	Section 4 of this report (Figures 7 to 10)
GitHub repository with scan analysis	This report, screenshots and scan analysis (link in the table above)
Demo video showing a scan and findings	LinkedIn post (link in the table above)

1. Lab and Scope

All scanning was done only inside my isolated lab. The passive recon used example.com, a domain reserved for documentation and testing.

Item	Details
Attacker machine	MahaKali (Kali Linux) with Nmap 7.991 and Nessus Essentials
Target machine	Metasploitable2, IP 192.168.56.102
Network	VirtualBox Host-Only network (private, no route to the internet)
Nmap scans	22 October 2026, 02:38 to 03:03
Nessus scan	21 October 2026, 11:25 PM to 11:41 PM

2. Reconnaissance

2.1 Passive Recon

Passive recon collects public information without touching the target.

Tool	Command	What I found
Whois	whois example.com	Domain created on 1992-01-01. The record shows the Internet Assigned Numbers Authority (IANA) as the organisation.
Nslookup	nslookup example.com	Two IPv4 addresses (104.20.23.154 and 172.66.147.243) and one IPv6 address.
Nslookup (NS)	nslookup -type=NS example.com	Name servers elliott.ns.cloudflare.com and hera.ns.cloudflare.com.
Nslookup (MX)	nslookup -type=MX example.com	A "null" mail record (priority 0, no server), so the domain does not receive email.

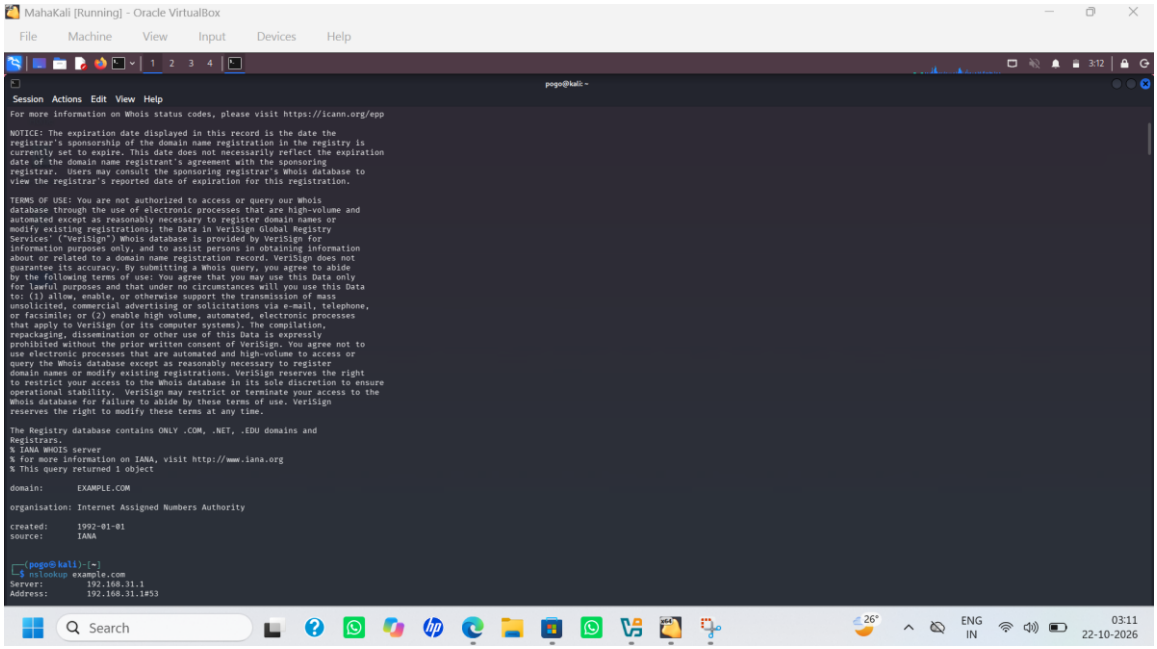


Figure 1: whois example.com result (domain created 1992-01-01, organisation IANA).

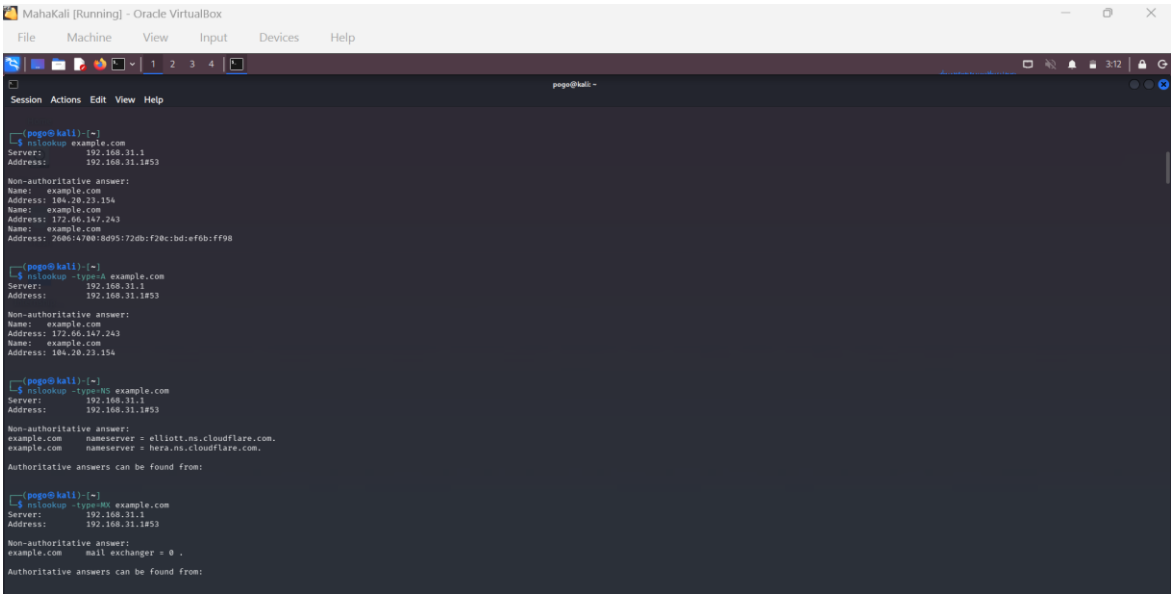


Figure 2: nslookup for A, NS and MX records of example.com.

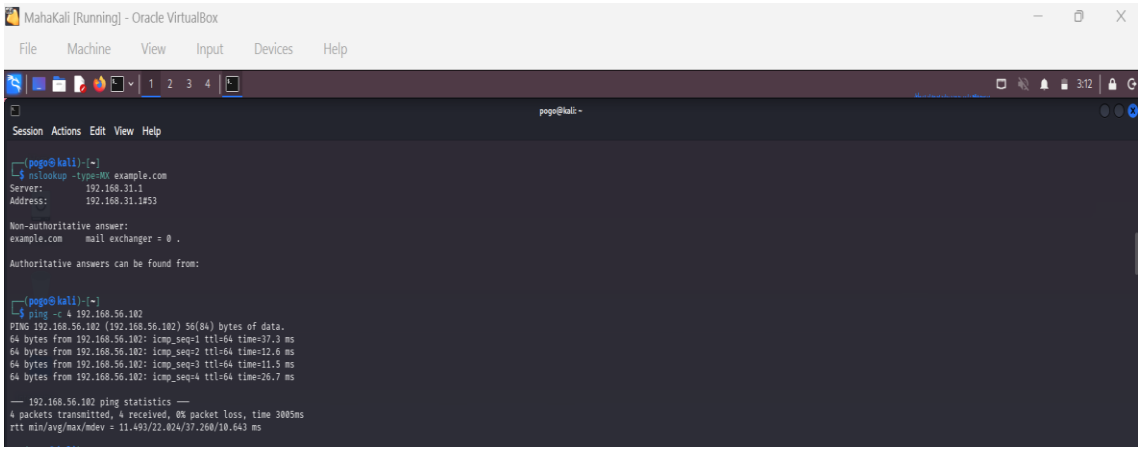


Figure 3: MX lookup, then ping to the target: 4 sent, 4 received, 0% packet loss.

2.2 Active Recon

Active recon talks directly to the target. I checked that the target is reachable and then ran a ping sweep of the lab network.

- **Ping test:** ping -c 4 192.168.56.102 got 4 replies, 0% loss, average 22.0 ms and TTL 64 (typical for Linux).
- **Ping sweep:** sudo nmap -sn 192.168.56.0/24 (Figure 4). The sweep listed many addresses as up. Only 192.168.56.102 was confirmed as Metasploitable2, by the ping reply and the full port scan, so all later scans used that address.
- **Banner grabbing:** the service banners and versions came from the -sV scan in Section 3.

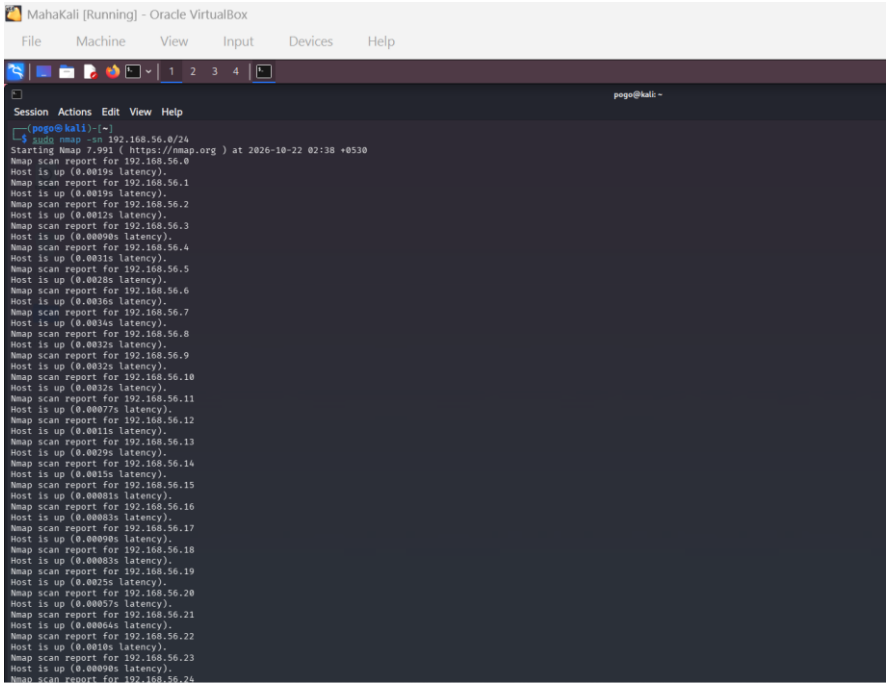


Figure 4: Ping sweep of 192.168.56.0/24 with sudo nmap -sn.

3. Nmap Scan Report

Target: 192.168.56.102 (Metasploitable2).

Scan	Command	Result
TCP scan with service versions	sudo nmap -sV 192.168.56.102	23 open TCP ports, 977 closed. Took 230.45 seconds. Run as root, so Nmap used a SYN scan.
UDP scan	sudo nmap -sU --top-ports 20 192.168.56.102	2 open (53 domain, 137 netbios-ns), 18 open|filtered. Took 3.98 seconds.
OS information	sudo nmap -O 192.168.56.102	Service Info from -sV shows hosts metasploitable.localdomain and irc.Metasploitable.LAN, OS Unix/Linux.

3.1 Open TCP Ports and Services

Port	Service	Version
21/tcp	ftp	vsftpd 2.3.4
22/tcp	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	telnet	Linux telnetd
25/tcp	smtp	Postfix smtpd
53/tcp	domain	ISC BIND 9.4.2
80/tcp	http	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	rpcbind	2 (RPC #100000)
139/tcp	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp	exec	netkit-rsh rexecd
513/tcp	login	Not identified
514/tcp	shell	Netkit rshd
1099/tcp	java-rmi	GNU Classpath grmiregistry
1524/tcp	bindshell	Metasploitable root shell
2049/tcp	nfs	2-4 (RPC #100003)
2121/tcp	ccproxy-ftp	Not identified
3306/tcp	mysql	MySQL 5.0.51a-3ubuntu5
5432/tcp	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	vnc	VNC (protocol 3.3)
6000/tcp	X11	(access denied)
6667/tcp	irc	UnrealIRCd
8009/tcp	ajp13	Apache Jserv (Protocol v1.3)
8180/tcp	http	Apache Tomcat/Coyote JSP engine 1.1

3.2 Observations

- **Plain-text services:** telnet (23), FTP (21) and rexec/rlogin/rsh (512 to 514) send data without encryption.
- **Port 1524:** Nmap labels it "Metasploitable root shell", so a root shell is listening on the network. This is the highest risk in the scan.
- **Old software:** vsftpd 2.3.4, OpenSSH 4.7p1, Apache 2.2.8, ISC BIND 9.4.2 and MySQL 5.0.51a are very old versions. vsftpd 2.3.4 is known for a backdoor in some copies.

- **Exposed remote and database ports:** MySQL (3306), PostgreSQL (5432), VNC (5900) and X11 (6000) are reachable from the network.
- **UDP:** DNS (53) and NetBIOS name service (137) are open. "open|filtered" means Nmap could not tell if the port is open or blocked.

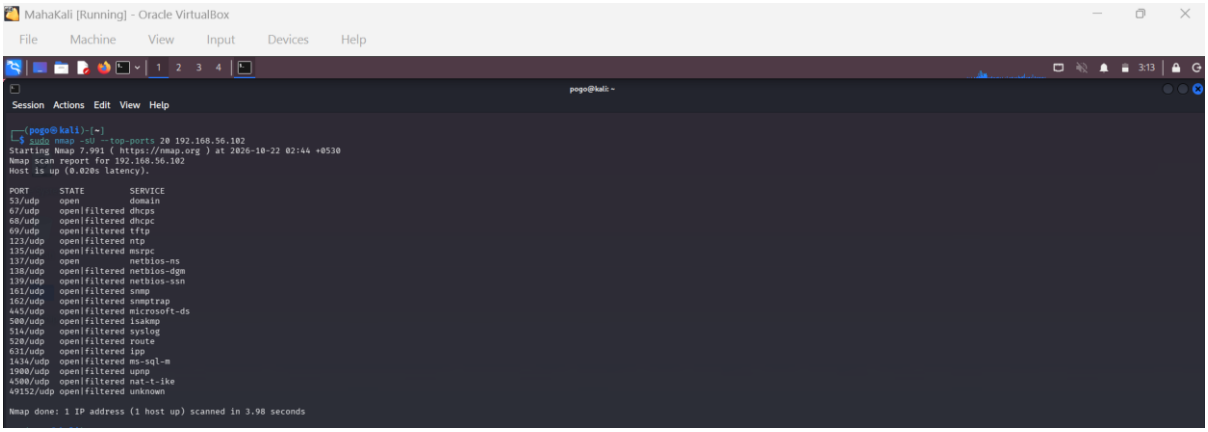


Figure 5: UDP scan: sudo nmap -sU --top-ports 20 192.168.56.102.

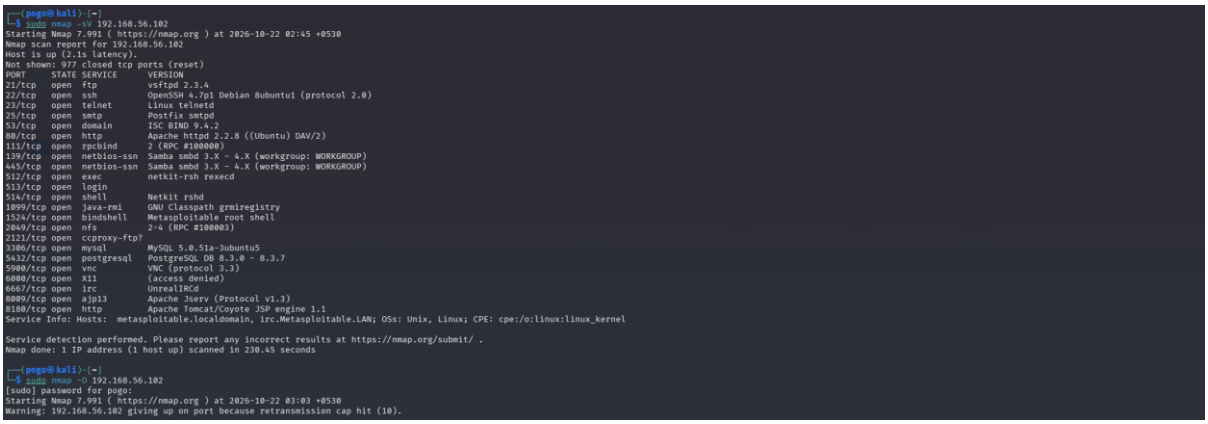


Figure 6: Service version scan: sudo nmap -sV 192.168.56.102 (OS detection started below it).

4. Vulnerability Scan Report (Nessus Essentials)

OpenVAS did not run properly in my lab, so I used Nessus Essentials, another vulnerability scanner, to scan the same target.

4.1 Scan Details

Item	Details
Scan name	Metasploitable2 vulnerability scan
Policy	Basic Network Scan
Target	192.168.56.102 (1 host)
Scanner	Local scanner on Kali, Nessus Essentials
Severity base	CVSS v3.0
Time	21 October 2026, 11:25 PM to 11:41 PM (16 minutes)
Authentication	Shows "Fail": the scan had no successful login, so it checked only what is visible from the network.

4.2 Results by Severity

Severity	Count
Critical	7
High	4
Medium	22
Low	8
Info	96

Nessus listed 50 vulnerability entries and 2 remediation actions for this host.

4.3 Main Findings

Severity	CVSS	Finding	Count	What it means
Critical	10.0	Canonical Ubuntu Linux SEoL (8.04.x)	1	The operating system is out of support, so it gets no security fixes.
Critical	10.0*	VNC Server 'password' Password	1	The VNC remote desktop uses a very weak password.
Critical	9.8	SSL Version 2 and 3 Protocol Detection	2	Old, broken SSL protocol versions are still enabled.
Critical	-	SSL (Multiple Issues)	3	A group of 2 related SSL issues bundled by Nessus.
High	7.5	Samba Badlock Vulnerability	1	A known Samba flaw. Fix: upgrade Samba.
Medium	6.5	TLS Version 1.0 Protocol Detection	2	TLS 1.0 is an old, weak version.
Medium	6.5	Unencrypted Telnet Server	1	Telnet sends logins in plain text.
Mixed	-	SSL (Multiple Issues)	28	A group of 15 related SSL issues.
Mixed	-	ISC Bind (Multiple Issues)	5	A group of 5 related BIND (DNS) issues.

4.4 Remediation Actions

Action	Vulns	Hosts
Upgrade ISC BIND to 9.11.22, 9.16.6, 9.17.4 or later (fixes a denial-of-service flaw).	3	1
Upgrade Samba to 4.2.11, 4.3.8, 4.4.2 or later (Samba Badlock).	0	1

Other fixes I would make: replace Ubuntu 8.04 with a supported system, set a strong VNC password, turn off SSL 2/3 and TLS 1.0, and use SSH instead of telnet.

4.5 Nmap and Nessus Agree

The two tools support each other. Nmap found VNC (5900), telnet (23), BIND 9.4.2 (53) and Samba (139, 445) open. Nessus then reported a weak VNC password, unencrypted telnet, BIND issues and the Samba Badlock flaw on the same services.

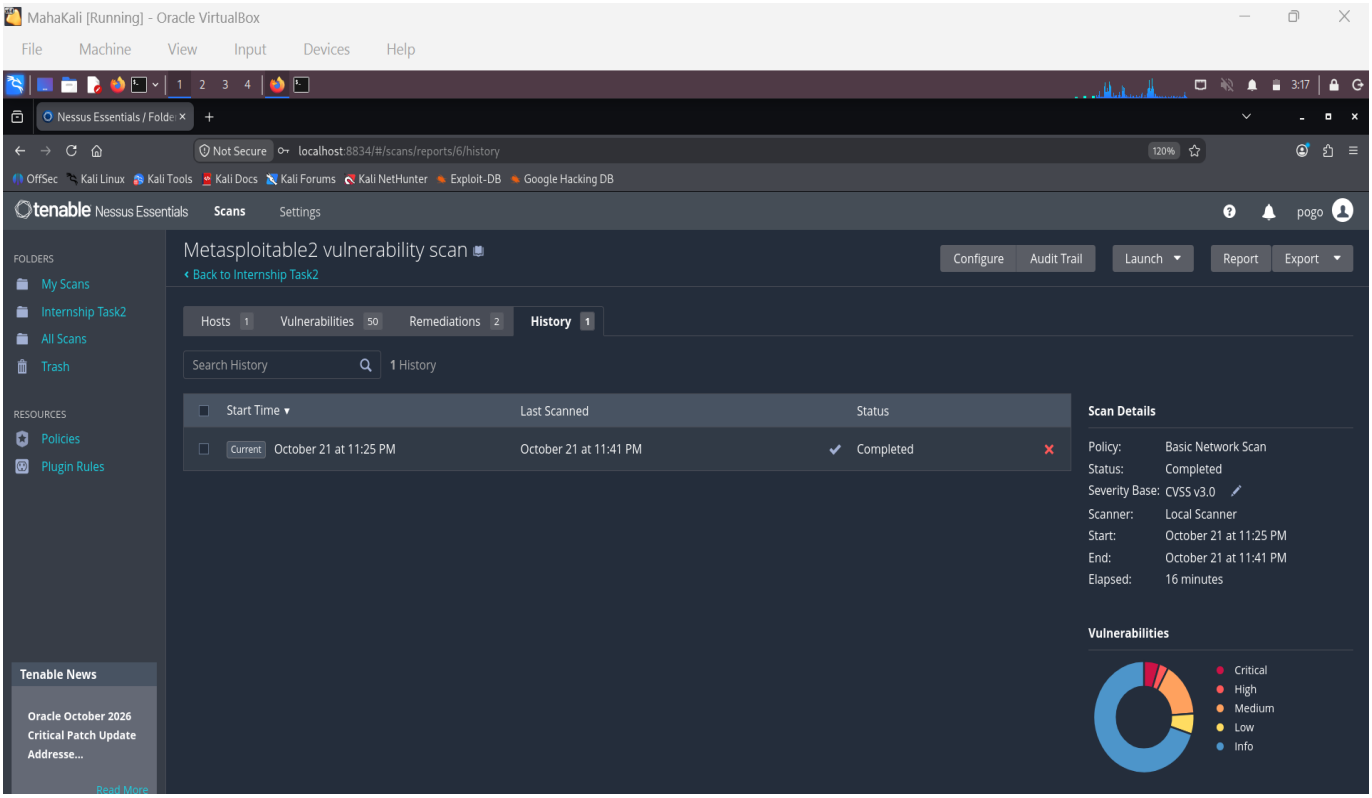


Figure 7: Nessus scan details: Basic Network Scan, completed in 16 minutes.

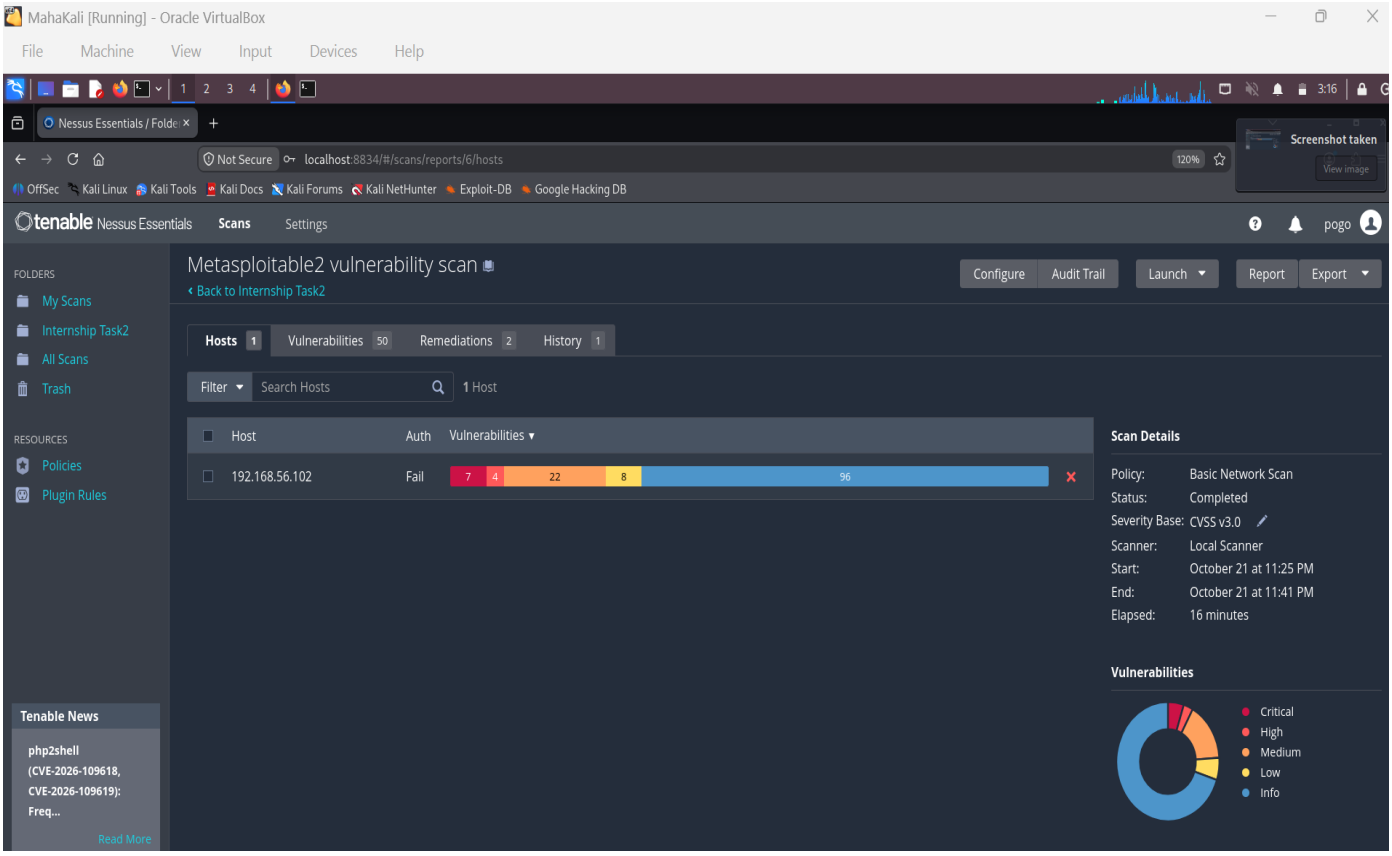


Figure 8: Nessus Hosts tab: 7 critical, 4 high, 22 medium, 8 low and 96 info results.

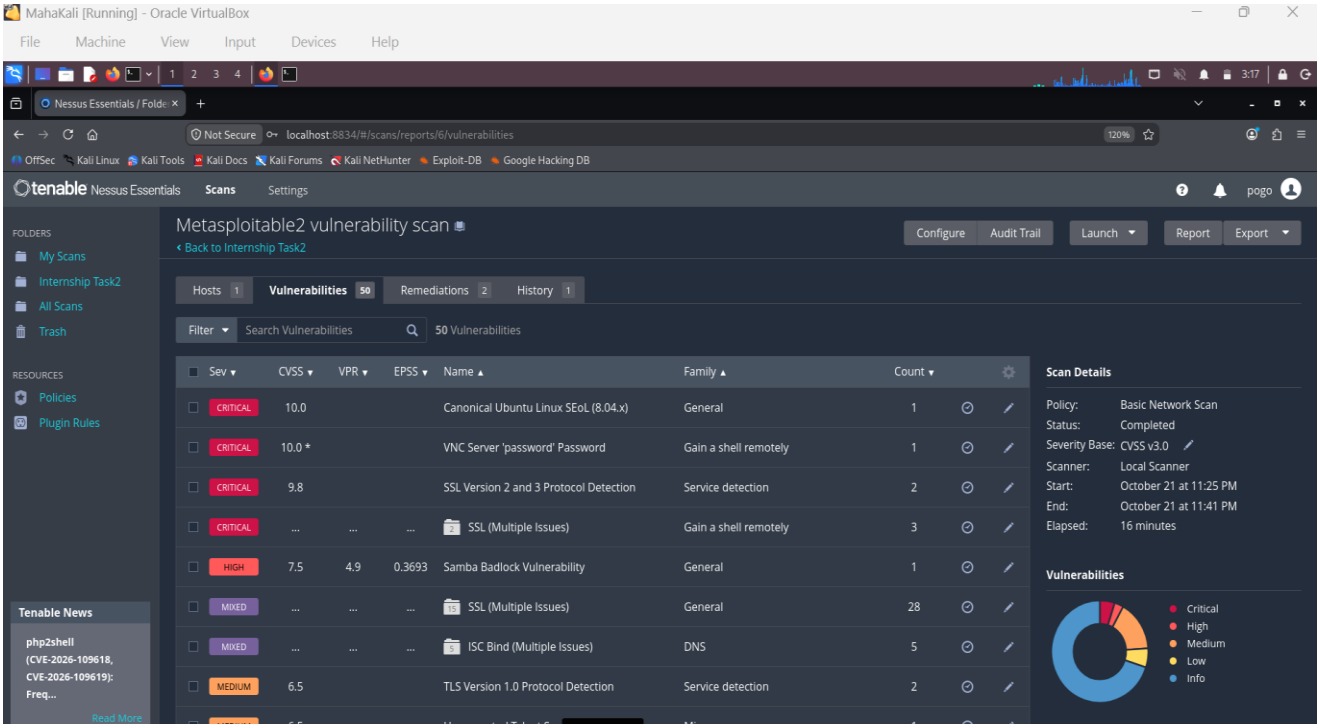


Figure 9: Nessus Vulnerabilities tab: critical and high findings.

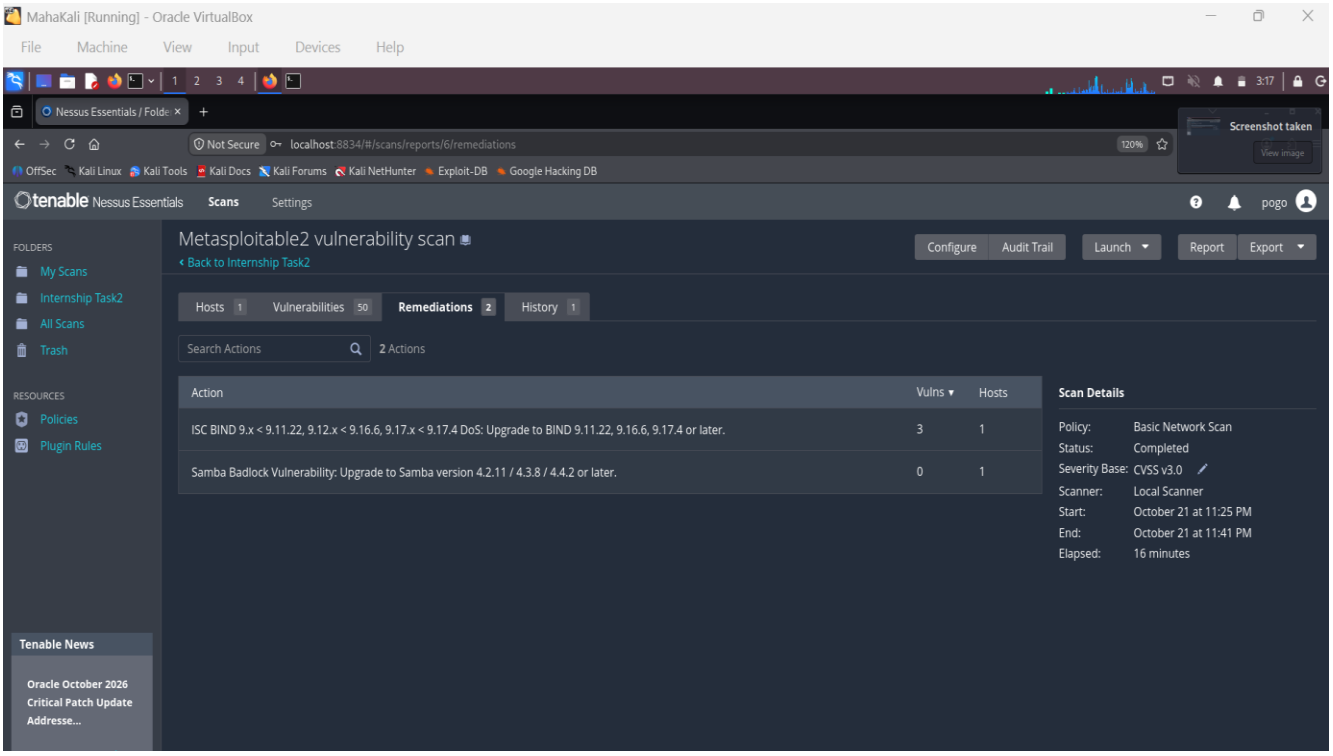


Figure 10: Nessus Remediations tab: BIND and Samba upgrades.

Conclusion

In Task 2 I collected public information about a domain, found 23 open TCP ports and 2 open UDP ports on my Metasploitable2 target, and ran a Nessus scan that reported 7 critical and 4 high results. The scans showed old software, plain-text services, weak passwords and open remote-access ports. I learned how recon, port scanning and vulnerability scanning work together, and which fixes matter first. All testing was done only in my own isolated lab.